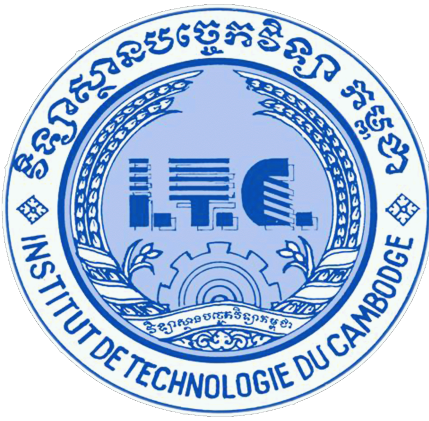




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Course: Networks System Design

Instructor: KUY Movsun

Assignment: Lab2

Due Date: November 04, 2025 (12:00 AM)

Activity 1 — Network Performance: Delay, Loss & Throughput

Part A. Manual Calculation

Scenario:

File size: 5 MB

Link speed: 10 Mbps

RTT: 50 ms

Packet loss: 2%

Packet size: 1500 bytes

1. Transmission Delay for One Packet

Formula:

$$\text{Transfer Time} = \frac{\text{File Size (bits)}}{R}$$

Where:

- File size = 5 MB = $5 \times 1024 \times 1024 \times 8 = 41,943,040$ bits

Calculation:

$$\text{Transfer Time} = \frac{41,943,040}{10,000,000} = 4.194 \text{ seconds}$$

2. Ideal Transfer Time for 5 MB File

Formula:

$$\text{Transfer Time} = \frac{\text{File Size (bits)}}{R}$$

Where:

- File size = 5 MB = $5 \times 1024 \times 1024 \times 8 = 41,943,040$ bits

Calculation:

$$\text{Transfer Time} = \frac{41,943,040}{10,000,000} = 4.194 \text{ seconds}$$

3. Compute the effective throughput with 2 % packet loss.

Formula:

$$\text{Effective Throughput} = R \times (1 - \text{loss rate})$$

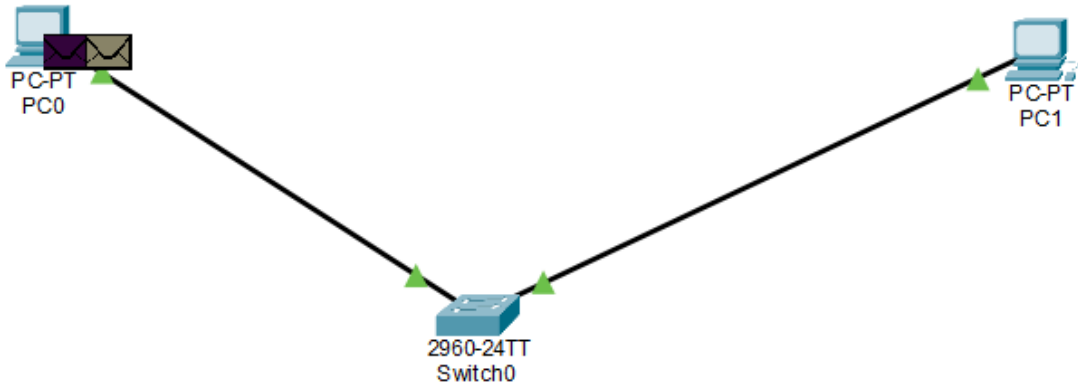
Calculation:

$$\text{Effective Throughput} = 10,000,000 \times (1 - 0.02) = 9,800,000 \text{ bps}$$

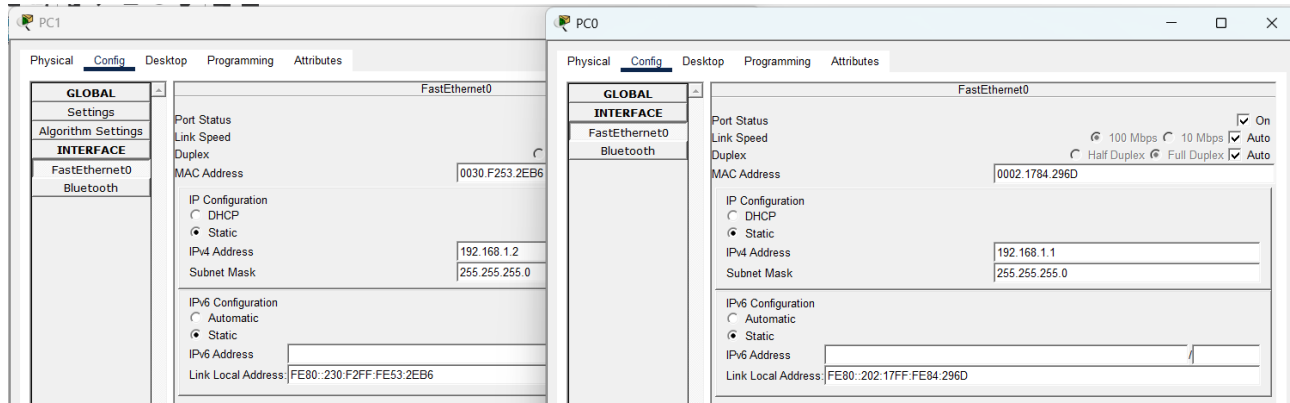
Metric	Value
Transmission Delay	1.2 ms
Ideal Transfer Time	~4.19 seconds
Effective Throughput	9.8 Mbps

Part B. Packet Tracer Measurement

- 1. Create a simple topology: PC0 — Switch — PC1 with straight-through cables.



2. ign IPs: PC0 192.168.1.1/24, PC1 192.168.1.2/24.



3. Test connectivity from PC0 using ping 192.168.1.2.

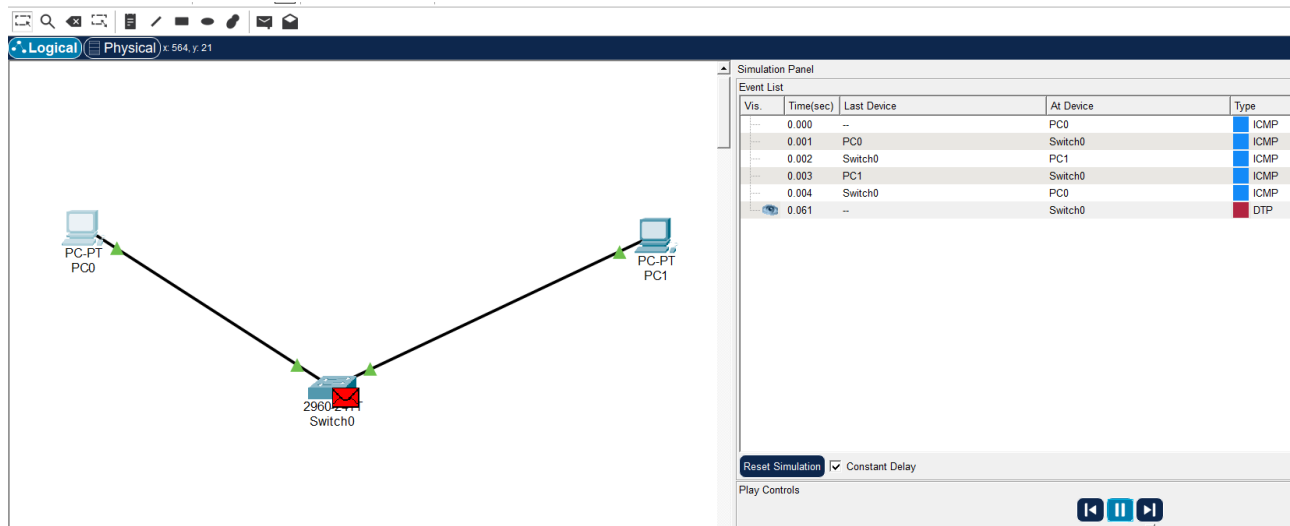
```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=6ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 6ms, Average = 4ms
```

- 4. Switch to Simulation Mode → use the Simple PDU (envelope icon) from PC0 → PC1.
- 5. Open the Event List and record the time when the Echo Request leaves PC0 and when the Echo Reply returns to PC0 (this is your RTT).
- 6. Assume each Simple PDU ≈ 600 bits (typical ICMP ping).



7. Calculate throughput using $\text{Throughput (bps)} = \text{Packet Size (bits)} / \text{RTT (seconds)}$.



Throughput Calculation

Formula:

$$\text{Throughput} = \frac{\text{Packet Size (bits)}}{\text{RTT (seconds)}}$$

Assuming packet size ≈ 600 bits:

$$\text{Throughput} = \frac{600}{0.004} = 150,000 \text{ bps}$$

Test	Packet Size (bits)	RTT (s)	Throughput (bps)	Loss (%)
1	600	0.004	150,000	0
2	600	0.004	150,000	0

Discussion Prompts

1. Which types of delay can you observe in Simulation Mode?

- **Transmission Delay:** Time to push bits onto the link (visible in packet timestamps).
- **Propagation Delay:** Time for bits to travel across the cable (minimal in short topologies).
- **Queueing Delay:** Not significant in simple setups, but can appear with multiple PDUs.
- **Processing Delay:** Time routers/switches take to inspect and forward packets.

2. How does sending multiple PDUs quickly affect queueing or packet loss?

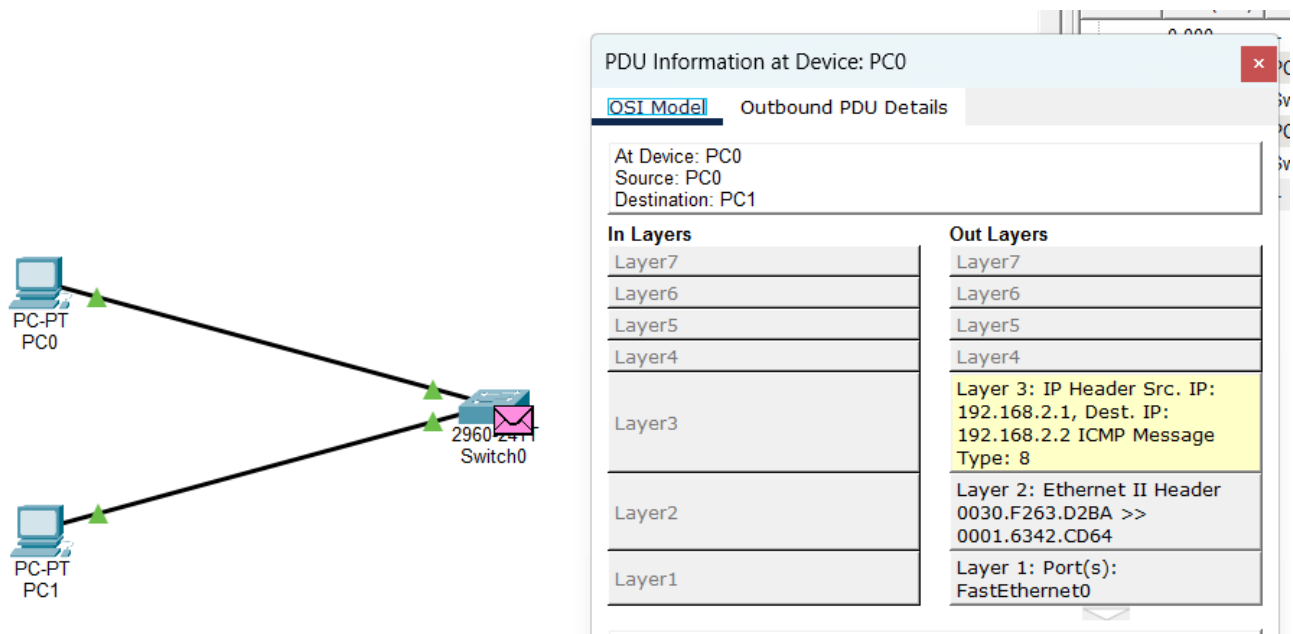
- Increases **queueing delay** as packets wait in buffer.
- May cause **packet loss** if buffer overflows (especially in congested or low-capacity links).
- Simulation shows delayed Echo Replies or dropped packets when PDUs are sent rapidly.

3. How does link speed influence throughput?

- Higher link speed = **faster transmission** = **higher potential throughput**.
- Throughput is limited by actual RTT and packet loss.
- In ideal conditions, throughput approaches link speed; in congested networks, it drops.

Activity 2 — Protocol Layers & Encapsulation

1. Use the same PC0 – Switch – PC1 topology in Simulation Mode.
2. Send a Simple PDU from PC0 → PC1.
3. In the Event List, click the packet → PDU Information.



4. Review of Outbound and Inbound PDU Details

Outbound PDU (PC0 → PC1)

- **Layer 3 (Network):** IP Header
 - Source IP: 192.168.2.1
 - Destination IP: 192.168.2.2
 - TTL is set here to control packet lifespan.
- **Layer 2 (Data Link):** Ethernet II Header
 - Source MAC: 0030.F263.D2BA
 - Destination MAC: 0001.6342.CD64
- **Layer 1 (Physical):**
 - Port: FastEthernet0

Inbound PDU (PC1 receives the packet)

- All headers are **removed** layer by layer:
 - **Layer 2:** Ethernet header is stripped to reveal IP packet.
 - **Layer 3:** IP header is removed to reveal ICMP message.
 - **Layer 4–7:** ICMP Echo Request is processed by the application.

5. Identify visible layers and headers (Ethernet/MAC, IP, ICMP).

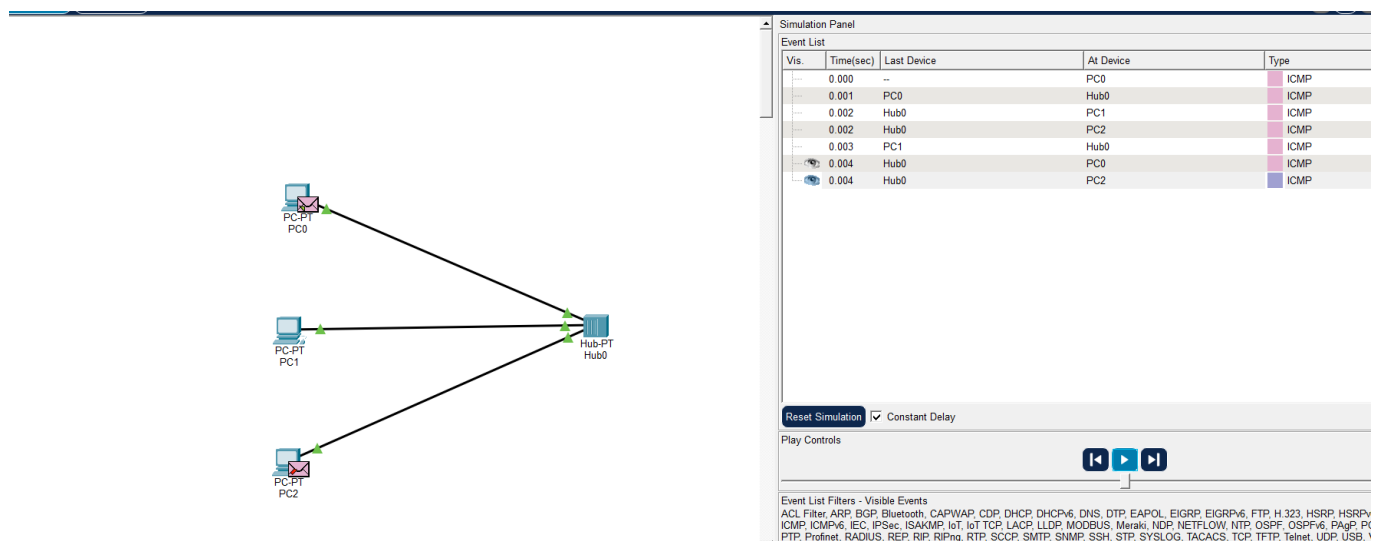
- **Data Link Layer (Ethernet/MAC)**
- Header: Ethernet II

- Source MAC: 0030.F263.D2BA
- Destination MAC: 0001.6342.CD64
- **Network Layer (IP)**
 - Header: IP
 - Source IP: 192.168.2.1
 - Destination IP: 192.168.2.2
 - TTL field is set here
- **Application Layer (ICMP)**
 - Header: ICMP
 - Type: 8 (Echo Request)

Activity 3 — Security Visibility: Hub vs Switch

Part A. Hub Network

1. Build 3 PCs (PC0, PC1, PC2) + 1 Hub.
2. Assign IPs: PC0 192.168.2.1, PC1 192.168.2.2, PC2 192.168.2.3
3. Switch to Simulation Mode → send Simple PDU from PC0 → PC1.

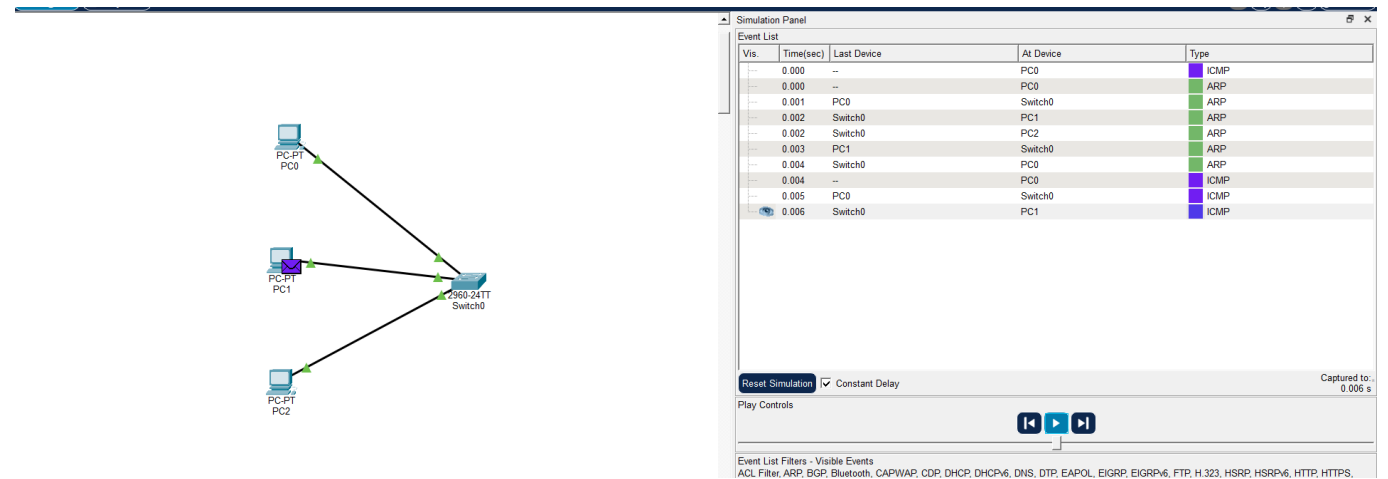


4. Observe whether PC2 receives a copy of the packet.

- **Topology:** PC1, PC2, PC3 connected to a hub
- **Observation:** When PC1 sends an ARP request, PC2 and PC3 also receive it.
- **Conclusion:** Hubs broadcast all traffic to every connected device. This makes network communication visible to unintended recipients, resulting in **low security**.

Part B. Switch Network

1. Replace the hub with a Switch and reconnect PCs (same IPs).
2. Repeat the Simple PDU test PC0 → PC1 and observe packet visibility at PC2.



- **Topology:** PC1, PC2, PC3 connected to a switch
- **IP Configuration:**
 - PC1: 192.168.2.1
 - PC2: 192.168.2.2
 - PC3: 192.168.2.3
- **Observation:** When PC1 sends a Simple PDU to PC2, PC3 does not receive a copy.
- **Conclusion:** Switches forward packets only to the intended recipient using MAC address lookup. This results in **high security** and prevents packet sniffing by other devices.

Hub vs Switch Security Comparison

Setup	Device	Packet Visibility at PC2	Security Level
Part A	Hub	Visible (broadcast)	Low
Part B	Switch	Not visible (unicast)	High